

A Unified Walking Model for Dimeric Motor Proteins

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ABSTRACT Dimeric motor proteins, kinesin-1, cytoplasmic dynein-1, and myosin-V, move stepwise along microtubules and actin filaments with a regular step size. The motors take backward as well as forward steps. The step ratio r and dwell time τ , which are the ratio of the number of backward steps to the number of forward steps and the time between consecutive steps, respectively, were observed to change with the load. To understand the movement of motor proteins, we constructed a unified and simple mathematical model to explain the load dependencies of r and of τ measured for the above three types of motors quantitatively. Our model consists of three states, and the forward and backward steps are represented by the cycles of transitions visiting different pairs of states among the three, implying that a backward step is not the reversal of a forward step. Each of r and τ is given by a simple expression containing two exponential functions. The experimental data for r and τ for dynein available in the literature are not sufficient for a quantitative analysis, which is in contrast to those for kinesin and myosin-V. We reanalyze the data to obtain r and τ of native dynein to make up the insufficient data to fit them to the model. Our model successfully describes the behavior of r and τ for all of the motors in a wide range of loads from large assisting loads to superstall loads.

INTRODUCTION

Most molecular motors associated with intracellular transport, such as kinesin-1 (conventional kinesin), cytoplasmic dynein-1, and myosin-V, are homodimers and move along linear tracks in a so-called hand-over-hand fashion using their head domains, just as we use our legs when we walk. Kinesin and cytoplasmic dynein move on a microtubule toward its plus and minus ends, respectively, whereas myosin-V translocates along an actin filament toward its plus (barbed) end (1,2). Extensive experimental work at the single-molecule level under different chemical and physical conditions with several techniques and mutants has revealed various properties of motor proteins (1,2). These data have inspired the theoretical modeling of the mechanism of how these biological motors work (3–5). In particular, experiments with loads applied to motor proteins are useful to elucidate how chemical energy is converted into mechanical work in these enzymes.

Theoretical models may be roughly grouped into the following two categories (5): physics-oriented ratchet-type models with continuous and discrete degrees of freedom (3,4,6–12) and chemical-kinetic models with discrete de-

grees of freedom (13–19). The former category is appealing in that it can provide the basic physical conditions required for directional motion to arise as a result of chemical transitions. The latter is useful for analyzing experimental data quantitatively because the mathematical treatment is easier and it is often the case that exact analytical solutions can be obtained. The chemical-kinetic models or physics-oriented models sometime contain a large number of parameters, and it is not easy to unambiguously determine the values of these parameters.

The purpose of our work is twofold. One is to propose a simple empirical model that can describe the stepping motion observed for kinesin, myosin-V, and dynein for wide ranges of loads and saturating ATP concentrations (20–28). The model proposed here has the advantage that it reformulated previous detailed models in terms of a few key parameters. This formulation makes it far more intuitive regarding how the ratio of the number of backward steps to that of forward steps (step ratio), the average time between consecutive steps (dwell time), and the velocity vary with the load. The other purpose is to reanalyze the data for these quantities for dynein obtained by Toba et al. (27) because there is not a sufficient amount of data for dynein in the literature to be used for quantitative analyses. We believe that, in the restricted condition of a high ATP concentration, our model provides a unified theoretical description of dimeric molecular motors in that motors of different types

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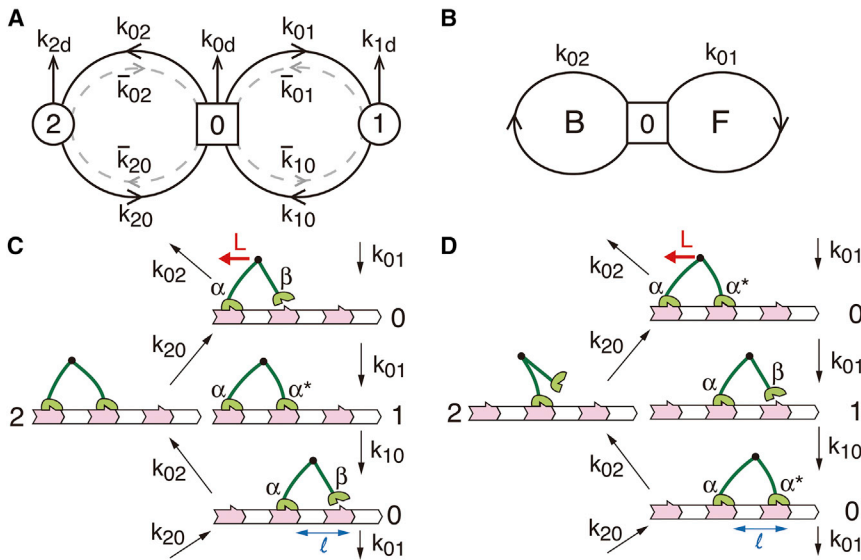


FIGURE 1 A three-state model for dimeric motor proteins for a saturated concentration of ATP. (A) A diagram representing three states (0, 1, and 2) and all possible transitions. The latter are represented by directed lines or arrows with k symbols indicating the rate constants. The cycles formed by the transitions k_{01} and k_{10} and by k_{02} and k_{20} , respectively, correspond to the forward and backward steps of the motor. The directed dashed lines with the symbols $\bar{k}_{ij}$ ($i, j = 0, 1, 2$) represent the reversals of the transitions k_{ij} , respectively. The vertical arrow with the symbol k_{id} ($i = 0, 1, 2$) indicates the detachment of the motor in state i from the track. (B) A simplified version of the three-state model obtained by neglecting the reverse transitions ($\bar{k}_{ij}$) and detachments (k_{id}) and assuming that k_{10} and k_{20} are much larger than k_{01} and k_{02} , respectively. The symbols “F” and “B” indicate that transitions k_{01} and k_{02} correspond to forward and backward transitions, respectively. (C and D) Two types of scenarios for a walking pattern on a linear track of period ℓ corresponding to the model in (A); the reverse transitions

and detachment processes are not shown. The motor primarily moves to the right in the absence of an applied load L . The state of the motor is labeled with a number (0, 1, and 2), whereas the chemical state of a head is labeled with a Greek letter (α , α^* , and β). In model C, the two-head-bound state (state 1) involved in a forward step ($0 \rightarrow 1 \rightarrow 0$) is different from that (state 2) in a backward step ($0 \rightarrow 2 \rightarrow 0$), whereas the one-head-bound state (state 0) is common in these steps. In model D, by contrast, the one-head-bound state (state 1) involved in a forward step ($0 \rightarrow 1 \rightarrow 0$) is different from that (state 2) in a backward step ($0 \rightarrow 2 \rightarrow 0$), whereas the two-head-bound state (state 0) is common. The chemical states of the heads in state 2 are not given explicitly because the chemical pathways of backward steps have not been clarified well by experiment. To see this figure in color, go online.

are represented by different values of parameters in a single model. The model is so simple that experimenters can readily use it to analyze their data, and it can easily be refined to take care of situations not considered explicitly in our work. For example, the model may be applied to describe the transport of cargo by multiple motors in cells. We try to give a clear and detailed exposition of the model for the benefit of readers that are not theory oriented.

METHODS

Three-state model

We construct a theoretical model to describe the stepping motion of dimeric motor proteins under conditions with high ATP concentrations. A globular head domain in each monomer catalyzes the ATP hydrolysis reaction, $\text{ATP} \rightarrow \text{ADP} + \text{P}_i$ (1,2). The chemical energy liberated by this reaction powers the motor. We propose a model represented by the diagram shown in Fig. 1 A, which consists of the following three states: 0, 1, and 2. In this model, a forward step corresponds to a cycle of transitions $0 \rightarrow 1 \rightarrow 0$ represented by solid lines, whereas a backward step takes the route $0 \rightarrow 2 \rightarrow 0$ (solid lines), which is different from the reverse of the forward step (transition $0 \rightarrow 1 \rightarrow 0$ represented by the dashed lines). Detachment of the motor from the track may occur in any state (vertical arrows in Fig. 1 A). In principle, the reverse transition is present for every transition (14–16,29); therefore, the backward step as the reverse of the forward step can also occur. However, there are several reasons why we consider a different route for a backward step. First, several authors proposed pathways for backward steps that are different from the reversal of the pathway for the forward step (12,22,23,26,30–33). Second, it was suggested from the energetics for kinesin revealed by the oxygen isotropic exchange method that the backward steps observed in *in vitro* motility experiments are not due to the simple reversal of the forward reaction (34). Third, because ATP is hydrolyzed in a forward step, it is necessary to supply ADP and P_i for

the reverse reaction to take place. However, in single-molecule experiments, P_i and ADP are not usually contained in the buffer (22–28); hence, the reverse transitions are unlikely to occur.

We introduce a simplifying assumption that the transitions from states 1 and 2 are so rapid that the lifetimes of these states are negligibly small compared with that of state 0 and adopt the following dependencies of the rate constants k_{01} and k_{02} on the load L :

$$\frac{1}{k_{01}} = \frac{1}{\kappa_1} + \frac{1}{\lambda_1} \exp\left(\frac{d_1 L}{k_B T}\right) \quad \text{and} \quad \frac{1}{k_{02}} = \frac{1}{\lambda_2} \exp\left(-\frac{d_2 L}{k_B T}\right), \quad (1)$$

where κ_1 , λ_n , and d_n ($n = 1, 2$) are positive parameters. We use the convention that a positive L means a force applied in the direction opposing motor movement and a negative L means a force assisting motor movement. Usually, a reaction rate is assumed to depend on the loads exponentially, as is the case for k_{02} in Eq. 1 (35). However, $1/k_{10}$ is given as the sum of a load-independent term and an exponential load-dependent term. Such an expression was proposed in (36) for RNA polymerase and then used in (22,37) to explain the load dependence of the dwell time for kinesin; that is, it is load independent for a low load and changes exponentially for a high load. An expression like the one for $1/k_{01}$ in Eq. 1 is usually expected for a process with two consecutive steps, one load dependent and the other not. However, we think that it represents a single-step event because an event with a single step can have the dependence of a rate constant on the load such that it is almost load independent at a low load and changes exponentially at a high load (38–40). The reason why we have assumed that the lifetimes of states 1 and 2 are negligibly small is that observed distribution of the dwell time is approximately single exponential (20,22,27,41). Therefore, it seems reasonable to assume in the first approximation that there is one rate-limiting process in a cycle. This simplification represents a significant difference of our model from previous models (12,22,23,26,30,32,33), in which a sequence of transitions in a single cycle or alternative cycles is involved for each forward and backward step or both forward and backward steps.

Detachment of the motor from the track does not occur frequently, at least for small loads for processive motors including kinesin, myosin-V, and dynein. Therefore, the effects of detachment are expected to be small, except on quantities directly related to detachment, such as the run length (the distance traveled by the motor until it is detached). If detachment and the reverse transitions are ignored and the simplifying assumption of the short lifetimes of states 1 and 2 is adopted, the kinetic diagram in Fig. 1 A is converted into the one in Fig. 1 B.

Although we mainly use this simplified model in the main text of this article, the analysis of the full model in Fig. 1 A is given in Section S4 of the [Supporting Materials and Methods](#), Table S1, and Fig. S3, and some of the results obtained there are referenced in the main text. The full model contains additional rate constants, and we need to introduce assumptions on the load dependence of these rate constants to analyze experimental data to be discussed in this article. These assumptions will bring a number of new parameters, and it will be difficult to determine the values of the whole parameters for all of kinesin, myosin-V, and dynein from currently available experimental data. Therefore, we did not use the full model for the analyses of the experimental data in our work.

For the three-state model in Fig. 1 A, two types of walking patterns on a polymer filament (track) shown in Fig. 1, C and D may be possible; we think that model C is suitable for kinesin and that model D is suitable for myosin-V and dynein, as explained in Section S1 of the [Supporting Materials and Methods](#) and Fig. S1. The chemical state of each head is distinguished by a Greek label (α , α^* , and β); plausible chemical states corresponding to these symbols for the three types of motors are explained in Section S1 of the [Supporting Materials and Methods](#) and Fig. S1. In model C, one head of the motor is bound to and the other is detached from the track in state 0, whereas both heads are bound to the track in states 1 and 2. The chemical state of the front (rear) head in state 1 is supposed to be different from that of the front (rear) head in state 2. In model D, both heads are bound to the track in state 0, whereas one head is bound to and the other is detached from the track in states 1 and 2. The chemical states of either or both of the bound and tethered heads in state 1 are supposed to be different from those in state 2.

Dwell time, step ratio, and velocity

The dwell time is defined as the period of time from a step to the next step or detachment. The average dwell times before a forward step, backward step, and detachment are denoted by τ_F , τ_B , and τ_D , respectively. The dependencies of these dwell times on the rate constants for the full model in Fig. 1 A are given in Eqs. S28, S30, and S31. They can be expressed as

$$\tau_i = \tau_C + \Delta\tau_i \quad (i = F, B, D), \quad (2)$$

where τ_C is common to the three dwell times, and $\Delta\tau_i$ represents the difference for different dwell times. Associated with the dwell times, the so-called splitting probabilities, p_F , p_B , and p_D , for the forward step, backward step, and detachment are defined as the probabilities that each of these events comes first after a step (they satisfy $p_F + p_B + p_D = 1$). The expressions for p_F and p_D are given in Eqs. S25 and S26. The step ratio r , defined as the ratio of the number of backward steps to that of forward steps per unit time is obtained as $r = p_B/p_F$. The run time τ_{run} and run length ℓ_{run} are defined as the period of time and the displacement of the motor, respectively, while the motor runs on the track until its dissociation from the track. They are given by Eqs. S20 and S22, from which the average velocity v of the motor is obtained as $v = \ell_{\text{run}}/\tau_{\text{run}}$ (12).

Under the conditions that the reverse transitions can be neglected and the lifetimes of states 1 and 2 are negligibly short, the expressions for the dwell times, step ratio, and velocity mentioned above can be simplified as follows ([Supporting Materials and Methods](#), Section S4.1). It turns out that $\Delta\tau_i$ for

$i = F, B$, and D in Eq. 2 are much smaller than τ_C , and we have the following:

$$\tau_F \approx \tau_B \approx \tau_D \approx \frac{1}{k_{01} + k_{02} + k_{0d}}. \quad (3)$$

The step ratio r and velocity v are now approximated by the following:

$$r \approx k_{02}/k_{01} \quad \text{and} \quad v \approx (k_{01} - k_{02})\ell. \quad (4)$$

Note that according to Eq. 4, detachment does not affect the step ratio and the average velocity in this approximation. By contrast, the dwell times given in Eq. 3 are affected by k_{0d} , but not by k_{1d} and k_{2d} . Considering the facts that quantitative experimental data on the detachment for analyzing the dwell time are available only for kinesin in a limited range of loads (2–8 pN) (22,42) and that the effect of detachment on the dwell time is expected to be small for processive motors for at least small loads, we will not consider the effects of detachment explicitly when analyzing the experimental data on the dwell time. Hence, we use the following expression,

$$\tau \approx \frac{1}{k_{01} + k_{02}}, \quad (5)$$

for the dwell time τ instead of Eq. 3. For kinesin, however, the effects of detachment will be discussed on the basis of the experimental data in (22,42).

It is remarked that Eqs. 4 and 5 can readily be obtained for the simplified model shown in Fig. 1 B. Because the dwell time is equal to the lifetime of state 0 irrespective of whether state 0 is terminated by a forward or backward step in this model, we obtain Eq. 5. Furthermore, Eq. 4 follows from the facts that the average numbers of forward and backward steps per unit time are given by k_{01} and k_{02} , respectively, and that the step size is ℓ .

RESULTS

Load dependence of the stepping properties

Substituting Eq. 1 into Eqs. 4 and 5, we obtain the load dependencies of the step ratio and dwell time. The results depend on the following five parameters:

$$\kappa_1, \lambda_1, d_1, \lambda_2, \text{ and } d_2. \quad (6)$$

To succinctly express the load dependencies of the step ratio, dwell time, and velocity, we introduce the constants a_1 , a_2 , L_1 , and L_2 defined by

$$a_1 \equiv \frac{d_1}{k_B T}, \quad a_2 \equiv \frac{d_2}{k_B T}, \quad L_1 \equiv \frac{1}{a_1} \ln \frac{\lambda_1}{\kappa_1}, \quad \text{and} \quad L_2 \equiv \frac{1}{a_2} \ln \frac{\kappa_1}{\lambda_2} \quad (7)$$

and functions $f_1(L)$ and $f_2(L)$ defined by

$$f_n(L) \equiv \exp[a_n(L - L_n)] \quad (n = 1, 2). \quad (8)$$

As will become apparent below, L_1 is the characteristic load at which the dwell time increases to approximately twice its unloaded value, whereas L_2 is the superstall characteristic load at which the dwell time becomes approximately equal

to the unloaded value; see Fig. 2 B below. With these parameters and functions, the rate constants can be expressed as $k_{01} = \kappa_1/(1+f_1)$ and $k_{02} = \kappa_1 f_2$. Now, substituting these expressions for k_{01} and k_{02} into Eqs. 4 and 5, we obtain the following:

$$r = (1+f_1)f_2, \quad \tau = \frac{1+f_1}{1+f_2+f_1 f_2} \frac{1}{\kappa_1},$$

$$\text{and } v = \frac{1-f_2-f_1 f_2}{1+f_1} \kappa_1 \ell.$$

The experimental data for the load dependencies of τ and r for kinesin, myosin-V, and dynein shown in Figs. 3, 4, and 5 were compared with the corresponding expressions in Eq. 9 to determine the values of the parameters listed in Eq. 6. From these values (Table 1), it is found that the inequality

$$L_2 - L_1 > 2/a_1 + 2/a_2 \quad (10)$$

is satisfied for all of these motors. As explained in Section S2 of the Supporting Materials and Methods and shown in Fig. S2, it can be shown that if this inequality holds, then f_2 is much smaller than unity or $f_1 f_2$; therefore, f_2 can be neglected in the denominator of the expression for τ and in the numerator of the expression for v to good approximation. Thus, by introducing the new constants a_{12} and L_{12} through

$$a_{12} \equiv a_1 + a_2, \quad a_{12} L_{12} = a_1 L_1 + a_2 L_2, \quad (11)$$

the load dependencies of r , τ , and v can explicitly be expressed as follows:

$$r = \exp[a_{12}(L - L_{12})] + \exp[a_2(L - L_2)], \quad (12)$$

$$\tau \approx \frac{1 + \exp[a_1(L - L_1)]}{1 + \exp[a_{12}(L - L_{12})]} \frac{1}{\kappa_1}, \quad \text{and} \quad (13)$$

$$v \approx \frac{1 - \exp[a_{12}(L - L_{12})]}{1 + \exp[a_1(L - L_1)]} \kappa_1 \ell. \quad (14)$$

We also find from the parameter values (see Table 1) that $a_1 > 0$, $a_2 > 0$ (i.e., $d_1 > 0$ and $d_2 > 0$), and $L_1 < L_2$; from these inequalities and the definitions of a_{12} and L_{12} in Eq. 11, we have the inequalities $a_1 < a_{12}$, $a_2 < a_{12}$, and $L_1 < L_{12} < L_2$.

Before we analyze the experimental data for r and τ by using Eqs. 12 and 13, the qualitative features of the load dependencies of r , τ , and v predicted by theory are explained. Noting that the ratio of the first term to the second term in Eq. 12 can be expressed as $\exp[a_1(L - L_1)]$ and regarding $e^{-2} \approx 0.135$ as a quantity that is much smaller than unity, we can approximate Eq. 12 (with an error less than 14%) as follows:

$$r \approx \begin{cases} \exp[a_2(L - L_2)] & (L < L_1 - 2/a_1) \\ \exp[a_{12}(L - L_{12})] & (L > L_1 + 2/a_1). \end{cases} \quad (15)$$

Therefore, when we plot the natural logarithm of r , $\ln r$, against L , it increases linearly with the slope a_2 for small L and then with the slope a_{12} for large L , as depicted in

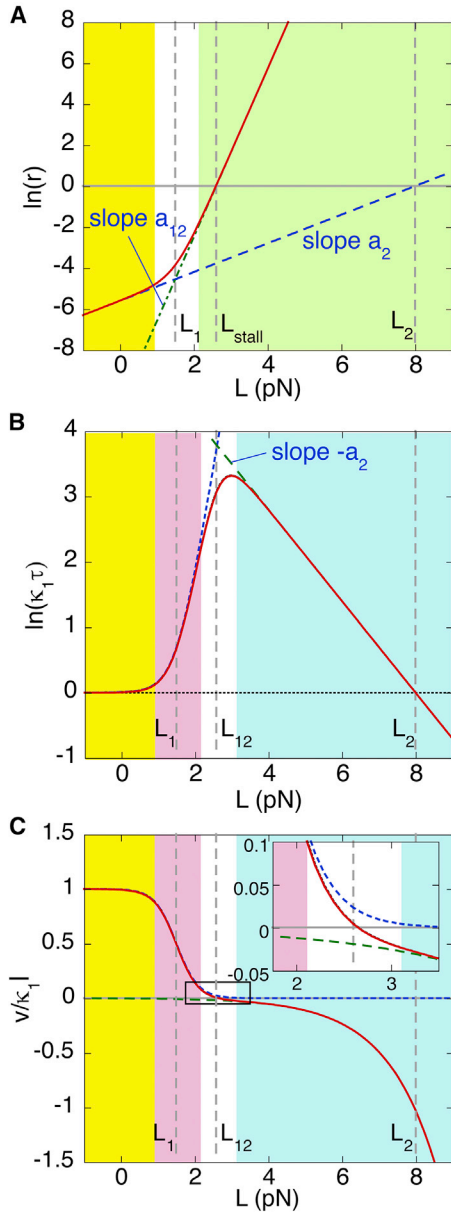


FIGURE 2 Theoretical predictions for the dependencies of the (A) step ratio r , (B) average dwell time τ , and (C) velocity v on the load L given by Eqs. 12, 13, and 14, respectively. The parameters $a_1 = 3.4 \text{ pN}^{-1}$, $a_2 = 0.7 \text{ pN}^{-1}$, $L_1 = 1.5 \text{ pN}$, and $L_2 = 8.0 \text{ pN}$ corresponding to myosin-V are used. The dashed and dash-dot lines in (A) represent the first and second expressions in Eq. 15, which are valid in the ranges of L shaded by yellow and green, respectively. The dotted, dashed, and long-dash lines in (B) and (C) represent the approximate expressions in Eqs. 16, 17, and 18, which are valid in the ranges of L indicated by yellow, purple and yellow, and blue shades, respectively. The inset of (C) shows an enlargement of the boxed region in the main panel.

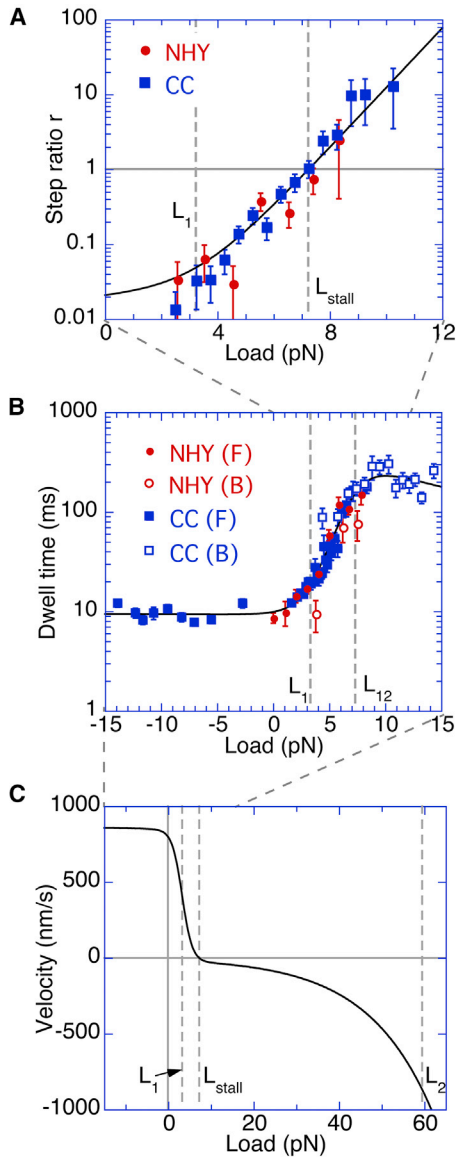


FIGURE 3 Dependencies of the (A) step ratio r , (B) dwell time τ , and (C) velocity v on the load L for kinesin. The experimental data (symbols) were compared with the theory (solid lines), Eq. 12 for r and Eq. 13 for τ with the parameters listed in Table 1. The vertical dashed lines indicate the values of L_1 , L_{stall} , L_2 , and L_{12} given in Table 1. The data indicated by NHY and CC are from (22) to (23), respectively. Kinesin from bovine brains was used in the experiment at 25°C by Nishiyama et al. (22), and *Drosophila* kinesin was used in the experiment at 23°C by Carter and Cross (23). The data on the dwell time from (23) were multiplied by a factor of 0.70 for the reason explained in the text. The dwell time data distinguished by the symbols “F” and “B” in parentheses are for those preceding forward and backward steps, respectively. The error bars represent the mean $\pm$ SE. To see this figure in color, go online.

Fig. 2 A. If $r = 1$, the motor does not move forward or backward on average. Hence, the value of L at which $r = 1$ corresponds to the stall load, which will be denoted by L_{stall} (the vertical dotted line labeled L_{stall} in Fig. 2 A). It can be shown that $L_{stall} \approx L_{12}$ under the condition in Eq. 10 (see Supporting Materials and Methods, Section S3); this is consistent with

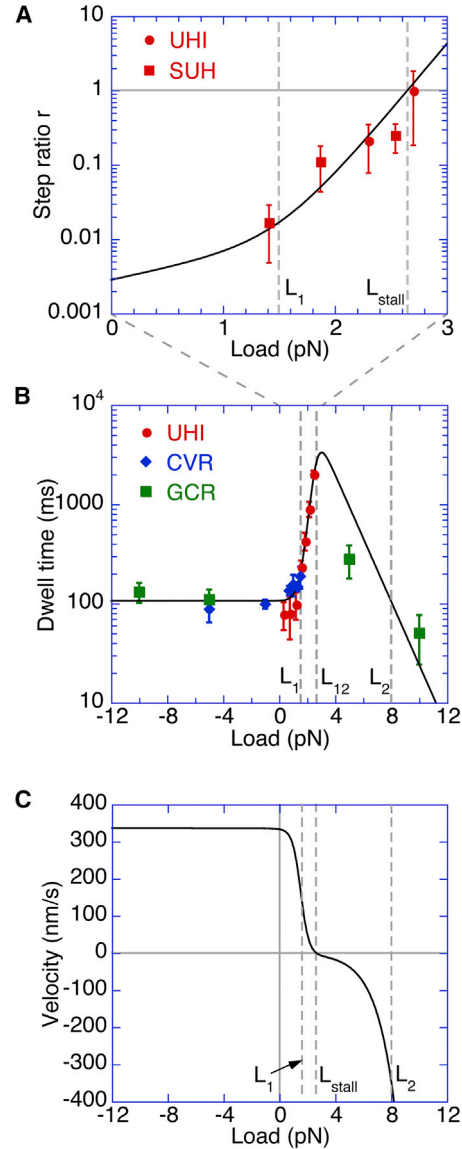


FIGURE 4 Dependencies of the (A) step ratio r , (B) dwell time τ , and (C) velocity v on the load L for myosin-V. The experimental data (symbols) were compared with the theory (solid lines), Eq. 12 for r and Eq. 13 for τ with the parameters listed in Table 1. The vertical dashed lines indicate the values of L_1 , L_{stall} , L_2 , and L_{12} given in Table 1. The data indicated by UHI, CVR, and GCR are from (24–26), respectively, and that indicated by SUH is from unpublished data by N. Sasaki, S. Uemura, H. Ishiwata, and H.H. presented at the 57th Annual Meeting of the Japan Society for Cell Biology in 2004. The step ratios were calculated by reanalyzing the data of UHI and SUH. Myosin-V used by all of these groups was extracted from chick brains, but the temperature and ATP concentration were somewhat different: 23–25°C and 1 mM ATP (24), 20–23°C and 2 mM ATP (25), and 20–23°C and 100 μ M ATP (26). The error bars represent the mean $\pm$ SE. To see this figure in color, go online.

the approximation in Eq. 15, which indicates that $v = 0$ at $L = L_{12}$.

Now, we consider the load dependencies of τ and v . As explained in Section S2 of the Supporting Materials and Methods, if $L < L_1 - 2/a_1$ is satisfied, then all of the

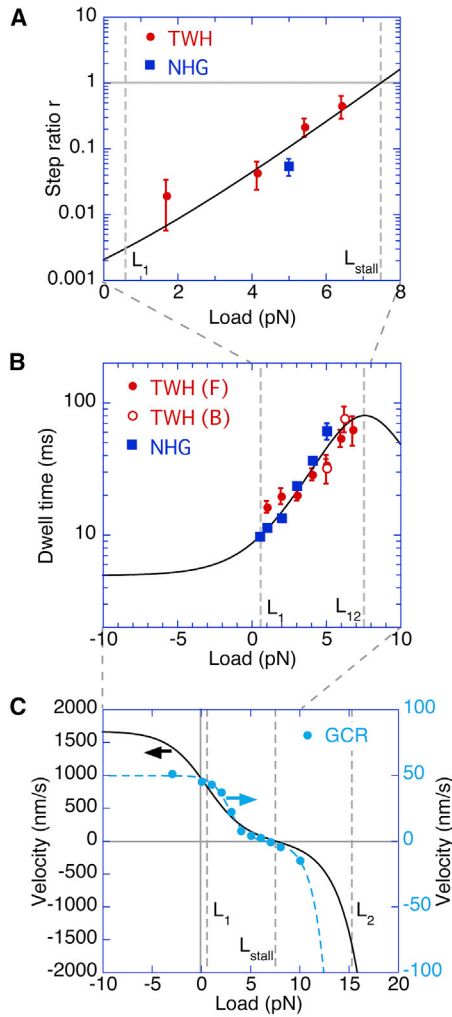


FIGURE 5 Dependencies of the (A) step ratio r , (B) dwell time τ , and (C) velocity v on the load L for cytoplasmic dynein. The experimental data (symbols) for r and τ were compared with the theory (solid lines), Eqs. 12 and 13, respectively, with the parameters listed in Table 1. The experimental data for v were fit to the theory (dashed line) with a different set of parameters (see the text for details). The vertical dashed lines indicate the values of L_1 , L_{stall} , L_2 , and L_{12} given in Table 1. The data indicated by TWH (reanalyzed in this work), NHG, and GCR are from (27,28) to (44), respectively. The data on the dwell time from (28) were multiplied by a factor of 0.74 for the reason explained in the text. The dwell time data distinguished by the symbols “F” and “B” in parentheses are for those preceding forward and backward steps, respectively. The error bars represent the mean $\pm$ SE. To see this figure in color, go online.

exponential functions in Eqs. 13 and 14 can be neglected under the condition that $L_1 < L_2$, which is the case for the three types of motors, as we have explained. Therefore, in this situation, we have a load-independent dwell time and velocity:

$$\tau \approx 1/\kappa_1 \text{ and } v \approx \ell/\tau \text{ for } L < L_1 - 2/a_1 \quad (16)$$

(see the domains colored yellow in the graphs in Fig. 2, B and C). This means that, in this regime of L , the forward-

TABLE 1 Model Parameters

Parameter	Kinesin	Myosin-V	Dynein
κ_1 (s^{-1})	110 ± 10	9.4 ± 0.6	203 ± 41
λ_1 (10^3 s^{-1})	1.7 ± 0.3	1.6 ± 0.9	0.27 ± 0.13
d_1 (nm)	3.5 ± 0.1	14 ± 1	2.1 ± 0.2
λ_2 (s^{-1})	2.1 ± 0.3	0.027 ± 0.009	0.24 ± 0.19
d_2 (nm)	0.27 ± 0.05	3.0 ± 0.3	1.8 ± 0.59
L_1 (pN)	3.2 ± 0.2	1.5 ± 0.2	0.58 ± 0.84
L_{12} (pN)	7.3 ± 0.3	2.7 ± 0.2	7.5 ± 1.5
L_2 (pN)	59 ± 13	8.0 ± 0.9	15 ± 5
L_{stall} (pN)	7.2 ± 0.3	2.7 ± 0.2	7.5 ± 1.5
α	0.70 ± 0.04	1.0	0.74 ± 0.21

The model parameters determined by least-squares fittings of Eqs. 12 and 13 to the experimental data for the step ratio and dwell time, respectively, are listed (fitting parameter $\pm$ SE of the fitting). The temperature was set at 25°C ($k_B T = 4.12 \text{ pN nm}$) in this analysis. The listed values of L_1 , L_{12} , and L_2 defined in Eqs. 7 and 11 and of the stall load L_{stall} (the load for which $r = 1$) were calculated by using the parameters thus determined. The parameter α is introduced to rescale the dwell time observed by some groups so that the data from different groups can be fit by a common set of model parameters: the data for kinesin in (23) and the data for dynein in (28) are multiplied by α listed in this table. Such calibration was not carried out for myosin-V ($\alpha = 1$).

stepping rate is dominated by the first term of $1/k_{01}$ in Eq. 1, and backward steps rarely occur.

As the load increases beyond the yellow regions in Fig. 2, B and C, the contribution from the exponential term $\exp[a_1(L - L_1)]$ in each of Eqs. 13 and 14 becomes appreciable, whereas the other exponential term in each of these equations is still negligible as long as $L < L_{12} - 2/a_{12}$. Therefore, we have the following:

$$\tau \approx \frac{1 + \exp[a_1(L - L_1)]}{\kappa_1} \text{ and } v \approx \frac{\ell}{\tau} \text{ for } L < L_{12} - 2/a_{12}. \quad (17)$$

In this situation, the dwell time increases, and the velocity decreases with increasing L ; τ is doubled, and v is halved at $L = L_1$ (the purple regions in Fig. 2, B and C).

After further increasing L , we enter the region in which the condition $\exp[a_{12}(L - L_{12})] \gg 1$ is satisfied. In this situation, the inequality $\exp[a_1(L - L_1)] \gg 1$ also holds under the condition in Eq. 10 (Supporting Materials and Methods, Section S2). Hence, noting that $a_1(L - L_1) - a_{12}(L - L_{12}) = a_2(L_2 - L)$, we obtain the following:

$$\tau \approx \frac{\exp[a_2(L_2 - L)]}{\kappa_1} = \frac{1}{\lambda_2} \exp\left(-\frac{d_2 L}{k_B T}\right) \quad (18)$$

$$\text{and } v \approx -\frac{\ell}{\tau} \text{ for } L > L_{12} + 2/a_{12},$$

which indicates that forward steps hardly occur in this range of loads. These exponential behaviors of τ and v can be seen in the blue regions in Fig. 2, B and C. Note that, at $L = L_2$, the dwell time of the backward step, Eq. 18, for superstall loads is equal to the dwell time of the forward step, Eq. 16, for assisting loads.

Analysis of kinesin data

Our theory was applied to analyze the data on the step ratio and dwell time for kinesin measured in (22) for 1 mM ATP and in (23) for 2 mM ATP (Fig. 3). In carrying out the analysis, we noticed that the dwell time obtained in (23) is somewhat larger than that in (22) (see Fig. S4), although the results for the step ratio by the two groups are similar. This difference in the dwell time is likely to originate from the differences in species and temperatures in the experiments (22,23). Therefore, we assumed in the analysis that all of the rate constants (κ_1 , λ_1 , and λ_2) associated with the kinesin used in (23) are common multiples of the corresponding rate constants associated with the kinesin used in (22). For this reason, we multiplied the dwell time obtained in (23) by a numerical factor α to describe their data and the data in (22) with a common set of model parameters.

The five parameters given in Eq. 6 and α described above were determined by least-squares fittings of Eqs. 12 and 13 to the data for the step ratio and dwell time, respectively. The parameters thus obtained are listed in Table 1. It turned out, as shown in Fig. 3, A and B, that the data obtained by the two groups (22,23) could be described well by a single theoretical curve for each of the step ratio and dwell time.

We also present the theoretical results for the velocity in Fig. 3 C, although it can be obtained from the results for the step ratio and dwell time. In experiments, the velocity is easier to observe than the step ratio and dwell time. Therefore, it would be useful to show how the velocity depends on the load in our model.

The values of L_1 , L_{12} , L_2 , and L_{stall} calculated from the expressions in Eqs. 7, 11, and 12 with the model parameters determined above and $k_B T = 4.12$ pN nm ($T = 25^\circ \text{C}$) are also listed in Table 1. As we have seen in the analysis of the theoretical results, the dependence of the step ratio on L qualitatively changes around $L = L_1$, and that of the dwell time qualitatively changes around $L = L_1$ and $L = L_{12}$. The locations of L_1 , L_{12} , and L_{stall} are indicated by the vertical dashed lines in Fig. 3.

The dwell times before the forward steps and those before the backward steps were analyzed separately in both (22,23) (the *solid* and *open symbols* in Fig. 3 B represent the dwell times before the forward and backward steps, respectively). Our theory predicts that the average dwell time before the forward steps coincides with that before the backward steps, which seems consistent with the experiments by the two groups (22,23).

Analysis of myosin-V data

The experimental data on the step ratio in (24), in unpublished data (see the legend of Fig. 4), and on the dwell time were analyzed (Fig. 4) in a similar manner as the case of kinesin described above. From the data on the velocity

in (25,26), the dwell time was estimated as the velocity divided by the experimentally measured step size of 36 nm (25). We introduced α to scale the dwell times observed in (25,26) (the conditions in these experiments are different from those in (24)) and obtained $\alpha = 0.98 \pm 0.14$; hence, we considered that $\alpha = 1.0$ within the uncertainty of the fitting. Therefore, we summarize the five model parameters listed in Eq. 6 determined by the least-squares fitting without such scaling in Table 1. The theoretical results for r and τ with these parameters agree well with the experiments from different groups.

Analysis of dynein data

We reanalyzed the data for porcine cytoplasmic dynein obtained at 1 mM ATP and 25°C in (27). The steps of dynein were detected on the basis of the bead displacement data using a step-finding algorithm (43); the noise in the data was reduced by a low-pass filter with a corner frequency of 2.0 kHz for the step-finding analysis (Fig. S5). The numbers of forward and backward steps were counted, and the load dependence of the dwell time was analyzed. We also collected the dwell time data for the forward and backward steps separately. The data for the step ratio and dwell time obtained in this manner are shown in Fig. 5, A and B.

We also obtained the data for these quantities from the results reported for the recombinant dynein motor domain from a rat in (28). In this experiment, the motor domain corresponding to the wild-type rat dynein (MD-WT) and the motor domain lacking the C-terminal region (MD- Δ CT) were investigated. The stall force of the MD-WT was about 1 pN, and that of the MD- Δ CT was about 6 pN, which is comparable to the stall force observed in (27). We analyzed the data for the MD- Δ CT obtained at 1 mM ATP and 25°C . The fraction of forward steps in 254 observed steps for $L = 5$ pN was reported to be 95, from which the data point for the step ratio in Fig. 5 A was obtained. The data for the dwell time shown in Fig. 5 B were obtained from the data for the force-velocity relation as follows. Let ϕ_+ and ϕ_- be the fractions of forward and backward steps ($\phi_+ + \phi_- = 1$). Assuming a constant step size of $l = 8.2$ nm, we can estimate the average dwell time τ from the velocity v as $\tau = (\phi_+ - \phi_-)l/v = (2\phi_+ - 1)l/v$. For the data of $L \approx 5$ pN, we use $\phi_+ = 0.95$, whereas we set $\phi_+ = 1$ for the data of smaller L .

The rate constants of recombinant dynein from a rat used in (28) can be different from those of the WT dynein from porcine brain used in (27). Therefore, the data on the dwell time in (28) were multiplied by a constant factor α to fit the data from different groups by a single theoretical curve. A least-squares fitting yielded the parameter values listed in Table 1.

The theoretical results and experimental data shown in Fig. 5, A and B are in good agreement. However, in contrast to the cases of kinesin and myosin-V, the experimental data

of the dwell time for assisting and superstall loads are lacking. The stepping motions for assisting and superstall loads were observed for dynein from yeast (44). The velocity measured for yeast dynein is plotted in Fig. 5 C and compared with our theory with the parameters $d_1 = 4.7$ nm, $d_2 = 3.2$ nm, $L_1 = 2.83$ pN, $L_2 = 11.56$ pN, and $\kappa_1 = 6.04$ s $^{-1}$, which are different from those of mammalian dynein listed in Table 1. We analyzed the data of the yeast dynein separately from those of the mammalian dynein because the yeast dynein exhibits multiple step sizes, and its velocity is much smaller than that of mammalian dynein.

DISCUSSION

We have proposed a model for dimeric motor proteins in which the inverse of the rate constant for the forward step is expressed as the sum of load-independent and load-dependent terms, whereas the expression for the rate constant for the backward step contains only an exponentially load-dependent term, as shown in Eq. 1. We were able to describe the dependencies of the step ratio and dwell time on a wide range of loads for different motor proteins, kinesin, myosin-V, and cytoplasmic dynein (Figs. 3, 4, and 5) by unified, simple analytical expressions (Eqs. 12 and 13) derived on the basis of an assumption about the rate constant. We obtained experimental data for the step ratio and the load dependence of the dwell time of native dynein, which enabled us to compare the three major motor proteins on equal footing.

Inspection of our model

The mechanochemical cycles for backward steps of kinesin, myosin-V, and dynein have not yet been clarified, in contrast to those for forward steps. Nevertheless, we assumed in our model that a backward step is not simply the reversal of a forward step. This property is shared by some models in the literature (12,22,23,26,30,32,33). Although some other models (14–16) assume that backward steps are simply the reversals of forward steps, reversals of forward steps are not likely to occur in *in vitro* motility experiments (34).

As described in Section S4 of the Supporting Materials and Methods, not only our simplified model in Fig. 1 B but also the full-version model in Fig. 1 A including the reverse transitions and detachment can be solved. If state 2 and the reactions associated with it are removed (Supporting Materials and Methods, Section S4.2), the full-version model reduces to the two-state reversible models studied in (14–16). Instead of considering such a two-state model, we considered the simplified model in Fig. 1 B for the reasons provided in the Methods. One of the conditions for our simplified model to be valid is that the reverse reactions can be neglected; this condition is satisfied if the products, ADP and P_i, of ATP hydrolysis are absent or their concentrations are negligibly small in the solution surrounding the motor. Because models without reverse transitions violate the

laws of thermodynamics, our simplified model cannot properly describe the thermodynamic efficiency of the motor. Another condition for simplification is that there is only one rate-limiting process in a cycle of a forward or backward stepping process. In such a model, the randomness parameter defined as $2D/v\ell$, where D is the diffusion coefficient of the motor, is always larger than unity if v is positive, which is in contrast to the experimental data for kinesin (21), which exhibit a randomness parameter less than unity for small loads. On the other hand, the single-rate-limiting approximation is in accordance with the single-exponential distributions of the dwell time observed for kinesin (20,22), myosin-V (41), and dynein (27). The dwell-time distribution in (22) obtained for loads in the range of 6–7 pN is consistent with the randomness parameter in (21), which is close to 1 in this range. However, the dwell-time distribution of single-exponential form in (20) was obtained for loads in the range of 3–4 pN, in which the randomness parameter in (21) is ~ 0.5 . It is not clear how the data of the dwell-time distribution can be reconciled with those of the randomness parameters for kinesin for small loads. For myosin-V, double-exponential distributions of the dwell time were observed (24,25,45). However, one of the rate constants is much smaller than the other, except for a small range of loads near the stall (in which the ratio of the two rate constants is about two). Therefore, our simplified model seems to be fine as a first approximation for myosin-V.

The load dependence of the detachment rate of myosin-V from an actin filament was measured for single-head myosin-V from mouse brain (46), and it was found that the detachment rate is described well by the expression for k_{01} given in Eq. 1 without the constant term $1/\kappa_1$. This constant term is essential for our model to explain the experimental fact that the dwell time is almost independent of the load for assisting loads. It is suggested that there can be a mechanism that brings a constant term to the detachment rate by forming a dimer.

Effects of detachment

We did not include the effects of the detachment of the motor from the track in the analysis of the experimental data presented above. Some data for detachment events of kinesin for loads of 2–8 pN have been reported (22,42). The dwell time τ_D between detachment and the step preceding it and the fractions p_F , p_B , and p_D of forward steps, backward steps, and detachment, respectively, after dwelling were measured in these studies. It was found that τ_D is almost the same as τ_F and τ_B , which is consistent with Eq. 3. Furthermore, the rates k_{01} , k_{02} , and k_{0d} can be calculated from these data as $k_{01} = p_F/\tau$, $k_{02} = p_B/\tau$, and $k_{0d} = p_D/\tau$; see Eqs. S43 and S45. It turned out that the detachment rate is roughly the same as the rate of backward steps (22,42) for loads of 2–8 pN. We have tentatively assumed that $k_{0d} = k_{02}$ holds for any load and analyzed

the experimental data for the dwell time and step ratio of kinesin shown in Fig. 3 using Eqs. 3 and 4. We find that these equations describe the data as well as the theoretical results presented in Fig. 3. The model parameters obtained from the fitting were not significantly different from those listed in Table 1, except for $d_2 = 0.01 \pm 0.01$ nm, which is smaller by 0.3 nm compared to that in Table 1. This difference affects the behavior of τ at large L , and experiments for τ at $L > 15$ pN are necessary to determine this parameter unambiguously.

The mean run length ℓ_{run} is related to the detachment rate k_{0d} as $\ell_{\text{run}} = (k_{01} - k_{02})\ell/k_{0d}$ (Eq. S41) for the motor type shown in Fig. 1 C if detachment occurs only when one head of the motor is bound to the track (i.e., $k_{1d} = k_{2d} = 0$) and the conditions for Eqs. 3 and 4 to be valid are satisfied. Using this equation for the run length with the model parameters determined as explained in the preceding paragraph, we obtain the load dependence of ℓ_{run} similar to those reported for kinesin in (37,47) for loads of 2–6 pN, and we have $\ell_{\text{run}} \sim 300$ nm at 2 pN, which is of the same order as ~ 500 nm in (37) and ~ 400 nm in (47). However, the run lengths observed in (47) for assisting loads are much smaller than our estimates, which implies that the relation $k_{0d} \approx k_{02}$ no longer holds in such a region.

It is interesting to note that we have a different expression for the run length for the motor type shown in Fig. 1 D from that for the type in Fig. 1 C if detachment only occurs when one head of the motor is bound to the track and the conditions for Eqs. 3 and 4 to be valid are satisfied. In particular, the expressions in the case of no load are $\ell_{\text{run}} \approx k_{01}\ell/k_{0d}$ for type C (Eq. S46), and $\ell_{\text{run}} \approx k_{10}\ell/k_{1d}$ (Eq. S47) for type D. The measured values of ℓ_{run} for myosin-V without a load are scattered over a wide range (48–51) from 0.7 to 2.4 μm . If we assume that $\ell_{\text{run}} = 1.5$ μm and use $\ell = 36$ nm for the step size of myosin-V, we obtain $k_{10}/k_{1d} \approx 42$.

Comparison with other theoretical models

As mentioned above, we obtained simple analytical expressions for the step ratio, dwell time, and velocity over a wide range of loads. Closed analytical expressions were obtained for the velocity for kinesin in (14), for the dwell time for kinesin in (16), and for the velocity and dwell time for myosin-V in (15). The velocities of kinesin and myosin-V for wider ranges of loads were analyzed on the basis of these kinetic theories with reversible transitions in (19). The behaviors of the velocity under a large assisting load and at high ATP concentrations predicted for kinesin and myosin-V in (19) are qualitatively different from those predicted by our model: the velocity decreases with increasing assisting load for kinesin and increases for myosin-V, whereas it converges to a constant value for both motors in our model. The velocities under superstall loads obtained in (19) are consistent with our results for myosin-V but do not agree with ours for kinesin (a nonmonotonic anomalous dependence on the

load was obtained in (19)). In these theories, a backward step is the reversal of a forward step, which does not seem to be realized in experiments without or with small amounts of ADP and P_i in the buffer solution. To justify the analysis by such theories, the authors (14) assumed that backward steps may be envisaged as resulting from ADP and P_i associated with kinesin and the microtubule in a successor complex for which the overall solution concentrations of ADP and P_i may have relatively little direct relevance. However, this assumption has not been verified experimentally. Therefore, these analyses will not be appropriate for the results in the in vitro experiments.

There were a few theoretical studies that explored wide ranges of loads numerically, in which backward steps are not reversals of forward steps. In (11), a detailed physics-oriented model for kinesin was constructed, in which the neck linkers were considered as entropic springs; the step ratio, dwell time, and velocity among others were calculated. The dwell time and velocity were independent of the load for both assisting and superstall loads, the latter property being in contradiction to our results. The physics-oriented polymer model for myosin-V studied in (12) predicted the dependence of the velocity on the load similar to that in (11). Kinetic models that consist of networks of chemomechanical cycles were used to analyze the properties, including the step ratio and velocity, of kinesin in (17) and myosin-V in (18). The load dependence of the velocity for myosin-V in (18) is qualitatively similar to our results. The dependence of the velocity on the load for kinesin in (17) qualitatively agrees with our results for the assisting load, but it shows a nonmonotonic dependence on the load for the superstall load, which is similar to the results in (19) but different from ours.

Step ratio

According to our model, the load dependence of the step ratio r changes its behavior around L_1 (Fig. 2 A). The step ratio increases exponentially with L for $L < L_1 - 2/a_1$ and more steeply increases exponentially for $L > L_1 + 2/a_1$ (Eq. 15). In previous work, the experimental data of r were fitted by a single exponential function for kinesin in (22,23) and for myosin-V in (52). To our knowledge, we are the first to predict that r should be described by two exponential functions depending on the region of L . However, the experimental data for kinesin, myosin-V, and dynein do not clearly show this qualitative change (Figs. 3 A, 4 A, and 5 A) because the data for $L < L_1$ are lacking. The preliminary results of the stepping ratio for kinesin for an assisting load supported our prediction (53).

The qualitative change in the load dependence of r around $L = L_1$ is due to the fact that the expression for the detachment rate k_{01} of the trailing head in Eq. 1 consists of a constant term and an exponential term, whereas the detachment rate k_{02} of the leading head is represented by a single

exponential function: the constant term in k_{01} is dominant for $L < L_1$, and the exponential term in k_{01} is dominant for $L > L_1$.

Dwell time

A qualitative change in the load dependence occurs around the characteristic load $L = L_1$ not only for the step ratio but also for the dwell time. The dwell time is independent of the load for $L < L_1 - 2/a_1$ (Eq. 16) because backward steps rarely occur for these loads, and the forward-stepping rate, k_{01} , is dominated by the constant term in Eq. 1. For kinesin and myosin-V, the experimental results (Figs. 3 B and 4 B) show that τ is almost independent of the load when it assists the movement of the motor ($L < 0$). In the case of dynein shown in Fig. 5 B, the data for assisting loads are lacking. However, the velocity data shown in Fig. 5 C support the prediction that the dwell time is independent of the load for $L < 0$ for dynein.

The dwell time increases steeply with the load for $L > L_1$ (Fig. 2 B) because the exponential term in the forward-stepping rate, k_{01} , given in Eq. 1 becomes dominant. In Figs. 3 B, 4 B, and 5 B, we observe such an increase, and it turns out that this behavior is described well by Eq. 17 up to a load of about $L_{12} - 2/a_{12}$.

As L increases beyond the characteristic load L_{12} , the fraction of forward steps becomes negligibly small for $L > L_{12} + 2/a_{12}$, in which the dwell time decreases exponentially, as given in Eq. 18. This decrease is very slow for kinesin (Fig. 3 B) and is much sharper for myosin-V and dynein (Figs. 4 B and 5, B and C). However, the experimental data are lacking for kinesin and dynein for superstall loads to confirm the prediction of the theory. Preliminary results for kinesin for superstall loads (53) indicate that they are consistent with our theory.

Characteristic distances

The parameters d_{01} and d_{02} that dictate the dependencies of the transition rates k_{01} and k_{02} , respectively, are usually called characteristic distances in the literature. We observe that the sum of these distances, $d_{01} + d_{02}$, for each motor listed in Table 1 is about half of the step size. The characteristic distance d_{02} associated with a backward step is much smaller than d_{01} associated with a forward step for kinesin; a similar observation was made in (22). For myosin-V, d_{02} is somewhat smaller than d_{01} . These parameters may be related to structural lengths such as the power-stroke distance and working distance (1). However, it is generally difficult to determine such a relation (54).

Perspective

The success of describing important experimental data on the motility of single motor proteins with quite different

structures by our unified walking model with a few key parameters suggests that these parameters capture the essence of the dynamic behavior of motor proteins. We envisage that the model as it is or with small modifications can be applied to other motor proteins.

Let us discuss such candidates of single molecular motors. RNA polymerase moves along DNA while carrying out transcription. Stepwise movement was observed for this protein (36,55). The forward stepping rate was found to consist of load-independent and load-dependent terms similar to that expressed in Eq. 1. Backward steps were also observed at high loads, but the step ratio was not reported (55). These data of RNA polymerase are not in conflict with the model presented here. Although RNA polymerase is not a dimeric motor like kinesin, the binding sites of polymerase to DNA will function as two heads. A second candidate, DNA polymerase, catalyzes DNA replication (56,57). The velocity (replication rate) was almost load independent for low loads and decreased with increasing load beyond 10 pN (56). The backward velocity increased exponentially with the load (57). These results are approximately consistent with our model.

Motor proteins work in a group in many cases in cells. Because of the simplicity of our model, it can be used to describe multimotor activities such as the transport of cargo and mitosis with some modifications. For example, we need to take account of the attachment rate of a motor detached from a track and the elasticity of motors that link a track and cargo or bridge different cytoskeletons. The applications of the unified walking model are not restricted to kinesin, myosin-V, and dynein but to a wider class of single motor proteins and multimotor activities.

SUPPORTING MATERIAL

Supporting Materials and Methods, five figures, and one table are available at [http://www.biophysj.org/biophysj/supplemental/S0006-3495\(18\)31156-1](http://www.biophysj.org/biophysj/supplemental/S0006-3495(18)31156-1).

AUTHOR CONTRIBUTIONS

K.S. and H.H. designed the research. K.S. performed the calculations. M.K. and H.H. analyzed the experimental data. K.S. and H.H. wrote the article.

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References (58–62) appear in the Supporting Material.

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